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Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil --Manuscript Draft--

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Abstract:	Body size can affect fecundity and access to mates in both sexes, yet field estimates of these selection components are uncommon for females and males in the same population. I sampled paired and unpaired adults of an undescribed flightless New Zealand weevil (<i>Lyperobates</i> sp. A) and examined body length, hind leg length, pairing status, female egg load, sexual dimorphism, and size assortment. Females were larger than males in both morphological traits, and longer females carried more eggs, consistent with fecundity selection favouring greater female size. In both sexes, body length was positively associated and hind leg length negatively associated with observed pairing status. The male analysis also supported nonlinear and correlational selection, with the highest predicted pairing probability among relatively large-bodied, short-legged males. A pooled comparison, however, did not clearly demonstrate that this correlational association differed between the sexes. Body length was not assortative within the 25 observed pairs. These results reveal size-dependent nonrandom pairing in both sexes and a fecundity-related advantage of female body size. They show that contemporary components of selection may contribute to female-biased dimorphism without mapping simply onto the magnitude of the observed sex differences.

Dear Dr Vasudeva,

Please find enclosed the revised version of manuscript ANBEH-S-26-00450, “Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil.”

I have carefully addressed the Editor’s and reviewers’ comments and substantially revised the manuscript. A detailed point-by-point response and the requested revision files accompany the submission.

Thank you for reconsidering the manuscript for publication in *Animal Behaviour*.

Sincerely,

Clint D. Kelly

Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil

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Data availability

Data are available through the OSF repository at

https://osf.io/khqgs/overview?view_only=728733529d06410ab17086f648a49a38.

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Conflict of Interest

I declare no conflict of interest.

Acknowledgements

I thank Steve Ward of the New Zealand Department of Conservation for logistical support while this work was conducted on Maud Island. I also thank Samuel Brown (The New Zealand Institute for Plant & Food Research Limited), Neil Birrell (University of Auckland), and Rich Leschen (New Zealand Arthropod Collection, Landcare Research Group) for help with taxonomic identification of my study species.

RESPONSE TO REVIEWERS

Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil

Manuscript ID: ANBEH-S-26-00450

Dear Dr Vasudeva and Reviewers,

Thank you for the careful and constructive evaluations. I have substantially revised the manuscript in response to your comments.

Page and line references refer to the clean revised manuscript.

Editor

comment: Reviewer #1 has some major suggestions surrounding the choice of either sex based on size when it comes to mating decisions. My own comments concern inbreeding and inbreeding load within the population because sampling was not extensive and recent evidence suggests that underlying genomic tensions can affect male fitness. Reviewer #1 also points out that the background and Discussion need more relevant information to support the rationale for using the weevils.

response: I have addressed these linked concerns throughout the revision. The Introduction now explains why this flightless, poorly known taxon is informative for estimating selection on both sexes in the wild and states what is and is not known about its mating behaviour (pages 5–6, lines 83–104). It also develops female choice, male choice, mutual choice, competition, shared genetic architecture, and inbreeding as relevant hypotheses (pages 3–4, lines 35–50). The Discussion returns to these mechanisms while distinguishing them from what can be inferred from the present cross-sectional data (pages 15–17, lines 304–352).

comment: As you revise your manuscript, please note that the journal requires that you address any animal welfare issues arising from your study within the Methods section, in a separate subsection headed Ethical Note. Address all ethical implications of the experimental design and procedures, including any procedures taken to minimise adverse impacts on the welfare of subjects or to enhance their welfare, for vertebrates and invertebrates alike.

response: I strengthened the separate Methods subsection headed “Ethical note” (pages 7–8, lines 132–143). It describes the absence of experimental behavioural manipulation, the measures used to minimize handling and unwanted interactions, individual housing, euthanasia before photography and dissection, why formal institutional approval was not required for this invertebrate study, collection and handling under permit from the New Zealand Department of Conservation, and compliance with applicable New Zealand regulations and the ASAB/ABS guidelines.

comment: Please also include reference to any Field Stations and/or Marine Laboratories in the Methods and Acknowledgements, as per the Guide for Authors.

response: The Methods now identify the New Zealand Department of Conservation field laboratory on Maud Island (page 7, lines 121–123). The Acknowledgements thank Steve Ward and the Department of Conservation for access to the laboratory and for logistical support (page 19, lines 368–375).

comment: Please prepare a detailed point-by-point response, provide highlighted and non-highlighted versions of the revised manuscript, upload the manuscript and figure source files, and supply a separate Highlights file.

response: This document provides a point-by-point response. The revision package includes the clean revised manuscript, the editable QMD and R source files, revised figure files, and a separate Highlights file. A version showing the revisions will also be included in the submission package.

Reviewer #1

comment: This is an interesting study that investigates patterns of non-random mating with regards to two morphological measures in both sexes in a wild population of an undescribed *Lyperobates* weevil. The study provides estimated selection gradients for body length and hind leg length based on pairing status and, for females, an additional fecundity estimate from egg load at collection. It is great to see studies looking at selection on both males and females in wild populations of nonmodel species. Overall, the questions are clear, the methods transparent, and the analyses and results straightforward, but the Abstract, Introduction, and Discussion require substantial reframing.

response: Thank you for this positive assessment and for the suggestions on framing. I have retained the central field study and analytical framework while revising the Abstract, Introduction, and Discussion to emphasize selection on both sexes, the value of a poorly known wild system, and the biological mechanisms that could generate size-dependent pairing.

Overall and major comments

comment: (1) The background explaining why the study was done and why the questions were asked is confusing. In addition to sexual size dimorphism, valuable reasons for such a study include examining selection pressures in both sexes, estimating selection in the wild and in nonmodel organisms, and increasing understanding of a species for which little is known. It is unclear why this undescribed species was chosen. Is anything known about mating duration, remating, and whether females can reject males? If not, make this clear.

response: I agree. The revised Introduction now explains that paired and unpaired adults of both sexes can be sampled under the same field conditions, that flightlessness makes body and leg morphology plausible contributors to mate searching and attachment, and that the taxon is poorly known (page 5, lines 83–92). It also states explicitly that no systematic observations were collected on mating duration, remating frequency, pair formation, female rejection or detachment, or male–male contests (pages 5–6, lines 92–97). I therefore describe the observations as consistent with, but not proof of, a scramble-competitive mating system. The final paragraph of the Introduction presents four explicit questions, including a direct comparison of the correlational morphology–pairing association between the sexes (page 6, lines 97–104).

comment: (2) Despite the main result of non-random mating with respect to body size in both males and females, which strongly suggests sexual selection for larger individuals, the Introduction and Discussion give this little attention. Female mate choice for larger males and, in particular, male mate choice for larger females should be mentioned. Males could benefit from mating with females that have a greater capacity for egg production, as body length appears to indicate.

response: I agree. The Introduction now explains that sexual selection on body size can operate in either sex. It discusses male choice for larger females when fecundity increases with size, female acceptance of larger males, and the possibility that size-dependent pairing reflects female choice, male choice, mutual choice, competition, or a combination of these processes. I added the curculionid *Diaprepes abbreviatus* as an example in which males preferred larger, more fecund females and females accepted larger males more readily (page 3, lines 35–41). The Discussion develops the corresponding hypotheses for *Lyperobates* while making clear that the data do not distinguish choice from encounter, mounting success, competition, attachment, or pair duration (page 15, lines 304–313).

comment: (3) I appreciate that cross-sectional studies have limitations because they provide a snapshot of the sampled population. However, because the Methods are clear, these limitations should be kept to one or two meaningful places, mainly in the Discussion and Methods. Otherwise, each result feels undermined immediately after it is reported. The study provides valuable estimates of selection on males and females and suggests that size matters; although conclusions cannot be definitive, suggestions can be made.

response: I agree with the concern and have reduced the repeated qualifications substantially. The Abstract now foregrounds the biological findings (page 2, lines 7–23), and the Results state the observed patterns directly. I retained only brief qualifications needed to define the response variables: egg load is not lifetime fecundity, and observed pairing status is not lifetime reproductive success. The broader limitations are consolidated in one Discussion paragraph (page 17, lines 343–352). The conclusion now emphasizes size-dependent nonrandom pairing, the association between female body length and egg load, and female-biased dimorphism before giving one concise qualification about the sex comparison (page 17, lines 353–359).

Abstract

comment: Lines 6–7: Consider adding more background before going directly to what was done. The reason for the study is missing. Before the sentence beginning on line 7, also state how the animals were collected, namely by sampling mating and non-mating individuals.

response: The Abstract has been rewritten. It now begins with the biological motivation and immediately states that paired and unpaired adults were sampled from the same wild population (page 2, lines 7–11).

comment: Lines 9–18: There are too many results for an Abstract. Consider removing some and retaining or summarizing only the main findings.

response: I shortened and consolidated the results in the Abstract. It now reports the central patterns in sexual dimorphism, female egg load, pairing-status selection, the male response surface and pooled sex comparison, and assortment, without the earlier statistical detail (page 2, lines 11–23).

comment: Lines 19–21: There is no need to mention this in the Abstract.

response: The extended qualification has been removed. The Abstract retains only the brief comparison needed to avoid interpreting the male-only statistical support as evidence of a demonstrated sex difference (page 2, lines 17–19).

comment: Lines 21–24: Why not reverse the arguments in this sentence so that the findings lead, followed by the qualification about whether the measured selection maintains or drove the evolution of sexual size dimorphism?

response: Revised as suggested. The final sentences now lead with size-dependent nonrandom pairing and the fecundity-related advantage of female body size, then note that contemporary selection does not map simply onto the magnitude of the existing sex differences (page 2, lines 20–23).

Introduction

comment: Line 33: Add that male mate choice for larger females may also contribute to the evolution or increase of dimorphism.

response: Added, together with an insect review and the *Diaprepes abbreviatus* example (page 3, lines 35–41). The possibility is also developed in the Discussion because female body length predicted egg load (page 15, lines 304–313).

comment: Lines 34–38: Consider explaining that many different selection processes can drive the evolution of sexual size dimorphism and that they can be difficult to separate.

response: The opening paragraph now distinguishes fecundity selection, sexual selection, viability selection, ecological divergence, developmental processes, and genetic constraints or correlations between the sexes. It also separates evolved dimorphism from contemporary phenotype–fitness associations (page 3, lines 24–34).

comment: Lines 40–42: In comparison with other animal groups such as mammals, female-biased size dimorphism is common among insects.

response: The Introduction now states that female-biased sexual size dimorphism is particularly common among insects and links this pattern to the frequent positive association between female size and fecundity (page 3, lines 27–31).

comment: After line 56: Add an example to illustrate the importance of multivariate selection.

response: I added a concrete example in which body length may influence competitive ability or persistence, whereas leg length may influence movement or the ability to remain attached. The text then explains how quadratic and correlational gradients can reveal high-fitness trait combinations (pages 4–5, lines 65–75).

comment: Lines 57–60: Consider removing this material or connecting it more clearly to examples and to what selection gradients allow researchers to identify.

response: The passage has been rewritten around the body–leg example. It now defines directional, quadratic, and correlational gradients in direct relation to the biological question (pages 4–5, lines 65–75).

comment: Lines 64–67: Assortative mating can also be a form of mutual mate choice.

response: Mutual choice is now identified explicitly both in the discussion of selection on body size (page 3, lines 35–41) and in the paragraph on assortative mating (page 5, lines 73–82).

Methods

comment: Lines 101–104: Instead of stating that the sample ratio should not be interpreted as an estimate of the population sex ratio, say that interpreting it as an estimate of the population or operational sex ratio should be treated with caution.

response: Rephrased as suggested. The Methods now state that the sample ratio should be interpreted cautiously and not as an estimate of the population or operational sex ratio (page 7, lines 114–123).

comment: Lines 109–111: Remove this because the same point is made in the Discussion.

response: I removed the repeated discussion-level interpretation. I retained one short statement defining egg load as a measure of current reproductive state rather than lifetime fecundity because this is needed to describe the response variable transparently (page 7, lines 124–128). The broader inferential limitation is treated in the Discussion (page 17, lines 343–352).

comment: Lines 131–135: Remove this from the Methods and present it in the Discussion as a limitation on inference.

response: I removed the extended interpretive discussion and consolidated the general limitations in the Discussion (page 17, lines 343–352). I retained a concise definition of observed pairing status and the processes that could produce it because those statements specify what the response variable represents and the assumption underlying the selection analysis (page 8, lines 145–151).

Results

comment: Lines 225–226: Remove the final part of the sentence and discuss it later as a limitation.

response: The repeated discussion has been removed. The Results now give the positive association between female body length and egg load directly, with one brief sentence defining the scope of the response variable; the general limitation is discussed later (page 13, lines 246–249; page 17, lines 343–352).

comment: Lines 233–236: The same applies here; discuss this material in the Discussion instead.

response: The detailed interpretation has been moved to the Discussion. The Results retain one concise sentence noting the possible processes underlying pairing status so that the observed association is not mislabeled as lifetime reproductive success (page 13, lines 250–258; page 17, lines 343–352).

comment: Lines 252–253: Use the term “allometry” when describing traits that scale differently as body length increases.

response: I added a formal static-allometry analysis rather than applying the term descriptively. The Methods now specify a log–log model of hind leg length as a function of body length, sex, and their interaction (page 8, lines 152–158). Hind leg length increased with body length, but

the allometric slope did not differ clearly between the sexes (body length $\times$ sex: $P = 0.223$; page 12, lines 233–245).

comment: Lines 254–264: Move the sexual-dimorphism section to the beginning of the Results.

response: Done. Sexual dimorphism is now the first Results subsection (page 12, lines 230–245).

comment: Lines 283–286: This repeats the qualification at the end of the Abstract.

response: The repeated qualification has been removed. The Results now report the direct pooled comparison between the sexes, and the broader inferential limitation is treated once in the Discussion (page 14, lines 269–275; page 17, lines 343–352).

Discussion

comment: Lines 302–303: Use “precopulatory success.”

response: The manuscript now describes observed pairing status as a measure of observed precopulatory success (page 17, line 346).

comment: Line 307: State more clearly that the pattern could reflect female mate choice for larger males.

response: The revised Discussion states this hypothesis explicitly and distinguishes it from male competition, mounting success, and remaining attached (page 15, lines 304–313).

comment: Lines 313–315: Add a couple of biological hypotheses and an example.

response: I added hypotheses involving movement, leverage, stability while mounted, and attachment endurance (page 15, lines 304–313). I also added *Callosobruchus maculatus* as a comparative example in which female kicking shortens copulation and reduces damage caused by male genitalia, while stating clearly that these behaviours and effects were not measured in *Lyperobates* (pages 15–16, lines 310–321).

Conclusion

comment: Lines 353–364: Phrase the conclusion with fewer limitations and emphasize that non-random pairing with respect to size was found, that males and females are sexually dimorphic, and that the results suggest that size matters.

response: The conclusion has been rewritten accordingly. It now leads with nonrandom pairing, female egg load, and female-biased dimorphism, then gives one concise qualification concerning the comparison of correlational selection between the sexes (page 17, lines 353–359).

Reviewer #2

comment: Using a flightless weevil species, the author investigates the direction of sexual size dimorphism and its underlying factors. The sexes did not exhibit opposing responses to the measured traits; the male nonlinear correlational response indicated that larger bodies combined with shorter legs predicted pairing likelihood. Larger females had more eggs, consistent with fecundity selection theory. Overall, this is a well-written paper with sex-specific insights into life-history traits.

response: Thank you for this positive assessment. I have retained the central analyses while clarifying the biological interpretation and adding a direct pooled comparison of the

correlational morphology–pairing association between the sexes. That comparison provided little evidence that the association differed between females and males ($z = -1.31$, $P = 0.190$; page 14, lines 269–275).

comment: Because the weevils were sampled from the wild, could inbreeding load be relevant to the population? A recent beetle study highlights tensions arising from the shared genome, in which even small amounts of inbreeding can be detrimental to male fitness. Could female-biased sexual size dimorphism drag the male genome towards a more feminized state? In the Introduction, after lines 50–53, could evidence from manipulations of inbreeding be used to illuminate misalignment between the sexes in fitness or fitness-related traits?

response: I agree that inbreeding is a useful hypothesis, but the present sampling provides no pedigree, relatedness, heterozygosity, or genomic-load estimates. I therefore added the mechanism without inferring that this population is inbred. The Introduction now discusses the shared genome, the strong intersexual autosomal genetic correlation for body size and sex-linked variance demonstrated in *Callosobruchus maculatus*, and experimental evidence that six generations of full-sib mating in *Tribolium castaneum* reduced male body size and several reproductive traits (pages 3–4, lines 42–50). The Discussion identifies inbreeding as an unmeasured potential modifier and explicitly states that it cannot be inferred from the restricted spatial sample (page 16, lines 322–336). I have avoided describing the males as “feminized” because the cited inbreeding experiment measured reductions in male morphology and reproductive performance rather than feminization itself.

comment: Line 91: Are geographical coordinates available for the New Zealand study site?

response: I added the precise location of the 20-m sampling transect on Te Pākeka/Maud Island (41.023594°S, 173.896832°E) to the Methods (page 6, lines 105–113).

comment: Line 112: In Fiji, was the segmented-line tool used?

response: Yes. The Methods now specify that the segmented-line tool in Fiji was used (page 7, lines 124–131).

comment: Lines 114–115: Were the leg measurements digital or made with callipers?

response: The measurements were digital. The Methods now state that body and hind leg length were measured digitally from scaled photographs using Fiji’s segmented-line tool, rather than with callipers (page 7, lines 124–131).

comment: Lines 301–303: In some other weevils, males harass females and cause reproductive-tract scarring and damage. Is anything known about the prolonged attachment behaviour in this species? Why did females not induce detachment, for example by kicking as in *Callosobruchus maculatus*?

response: I added a focused Discussion paragraph on pair duration and sexual conflict. It describes the *C. maculatus* example and states that attachment duration, kicking, induced detachment, genital morphology, and reproductive-tract scarring were not measured in *Lyperobates*. It also explains that phenotypes associated with prolonged attachment could be overrepresented among cross-sectionally observed pairs (pages 15–16, lines 310–321). This preserves the useful hypothesis without implying that genital damage or kicking occurs in the study species.

comment: Figure 4: Can the points within the contours be made clearer, perhaps by using colours that contrast with the background?

response: The male response surface, now Figure 5 following the corrected order of first appearance, has been redesigned. Unpaired males are shown as white circles and paired males as yellow triangles, both with black borders, against the coloured probability surface. A legend and revised caption identify the two categories.

Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil

Highlights

- Females exceeded males in both body and hind leg length.
- Larger females carried more eggs, consistent with fecundity selection.
- Body length was positively associated with pairing status in both sexes.
- Hind leg length was negatively associated with pairing status in both sexes.
- Large-bodied, short-legged males had the highest predicted pairing probability.

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Sexual size dimorphism and multivariate phenotypic selection in a flightless weevil

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Abstract

~~Sexual~~Body size dimorphism (SSD) can arise from sex differences in ~~fecundity, sexual and viability selection, but contemporary fitness measures do not necessarily reveal access to mates in both sexes, yet field estimates of these selection components are uncommon for females and males in the processes that produced dimorphism, same population.~~ I examined morphology, egg load, and pairing status in ~~sampled paired and unpaired adults of~~ an undescribed flightless New Zealand weevil (*Lyperobates* sp. A), ~~focusing on body and hind leg length. Female body length was positively associated with the number of mature A) and examined body length, hind leg length, pairing status, female egg load, sexual dimorphism, and size assortment. Females were larger than males in both morphological traits, and longer females carried more eggs present at dissection, consistent with fecundity selection favouring larger females, although egg load does not measure lifetime reproductive output. Directional selection gradients based on pairing status were qualitatively similar in greater female size. In both sexes,~~ body length was positively associated ~~with pairing status, whereas~~ and hind leg length ~~was~~ negatively associated with it. Thus, females and males did not experience opposing directional selection regimes ~~observed pairing status.~~ The male analysis also supported nonlinear and correlational selection, with the highest predicted pairing probability among ~~males combining relatively large bodies with bodied, short legs. However, a legged males. A~~ pooled analysis provided little evidence ~~comparison, however, did not clearly demonstrate that the body length $\times$ leg length this correlational~~ association differed between the sexes, ~~so this contrast should not be interpreted as male-specific correlational selection. Females were larger than males in both traits, and body~~ Body length was not assortative among within the 25 observed pairs. ~~Because pairing status was measured during a single sampling period, it may reflect encounter, mounting, or attachment~~

27 ~~duration rather than lifetime mating or reproductive success. These results document female-~~
28 ~~biased SSD and similar contemporary directional pairing status selection in both sexes, but do~~
29 ~~not establish that the measured selection components caused or currently maintain the~~
30 ~~dimorphism~~ These results reveal size-dependent nonrandom pairing in both sexes and a
31 fecundity-related advantage of female body size. They show that contemporary components of
32 selection may contribute to female-biased dimorphism without mapping simply onto the
33 magnitude of the observed sex differences.

34

Introduction

Sexual size dimorphism (SSD) is a widespread and evolutionarily significant feature of animal morphology (Darwin, 1871; Fairbairn et al., 2007). Its direction and magnitude can reflect sex-specific fecundity and sexual selection, as well as viability selection, ecological divergence, developmental processes, and genetic constraints or correlations between the sexes (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007). ~~In many species, male-biased SSD is associated with sexual selection arising from male-male competition or female choice (Andersson, 1994), whereas female-biased SSD~~Female-biased SSD is particularly common among insects and is often associated with fecundity selection because larger females can produce or carry more eggs (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017; Zhu et al., 2025). ~~Nevertheless, a~~ positive size-fecundity relationship ~~between female body length and fecundity, however, does not, on its own, by itself~~ demonstrate that fecundity selection produced SSD. Dimorphism is an evolved outcome of selection acting across generations, whereas estimates from ~~a single~~one population and ~~timesampling~~ period describe contemporary ~~phenotypic~~ associations with particular components of fitness.

Sexual selection on body size can also operate in either sex. When female fecundity increases with size, males may benefit by preferentially courting, mounting, or investing in larger females (Bonduriansky, 2001). In the curculionid beetle *Diaprepes abbreviatus*, for example, males preferred larger, more fecund females, females accepted large males more readily, and competition contributed to size-assortative mating (Harari et al., 1999). Nonrandom pairing by size may therefore reflect female choice, male choice, mutual choice, competition, or some combination of these processes.

The sexes nevertheless share most of their genome, which can constrain independent responses to sex-specific selection. Experimental work on the seed beetle *Callosobruchus maculatus* revealed a strong intersexual autosomal genetic correlation for body size, while sex-linked variance enabled SSD to respond to selection (Kaufmann et al., 2021). Inbreeding can further alter phenotype–fitness relationships by exposing deleterious recessive variants; for example, six generations of full-sib mating in *Tribolium castaneum* reduced male body size, testis size, sperm length, lifespan, and reproductive output (Vasudeva et al., 2025). Population structure, relatedness, and genetic load could consequently influence sex-specific morphology and fitness, although evaluating those mechanisms requires pedigree or genomic information.

When sexual selection favours increased male body length, as often occurs under male–male contest competition, male-biased SSD may evolve, or female-biased SSD may be reduced (Andersson, 1994; Emlen & Oring, 1977). Conversely, female-biased SSD has sometimes been attributed to weak sexual selection on males or to selection favouring smaller, more mobile males (Fairbairn et al., 2007; Ghiselin, 1974). Scramble-competition polygyny is frequently invoked in this context because male reproductive success can depend on locating and securing females rather than defending resources (Andersson, 1994; Moya-Laraño et al., 2002). Smaller males, or males with locomotor morphology that facilitates mate searching, have higher mating success in some systems, including Japanese beetles (*Popillia japonica*) and Cook Strait giant weta (*Deinacrida rugosa*) (Kelly et al., 2008; Kelly, 2020; Kelly & Gwynne, 2023). Comparative evidence, however, shows that larger males are also favoured in many scramble-competitive arthropods (Herberstein et al., 2017). The direction of selection on male morphology therefore cannot be inferred from a broad mating-system label alone, especially when mate searching and pair formation have not been observed directly.

81 Multivariate analyses are important because the association between one trait and a fitness
82 component can depend on other traits (Brodie et al., 1995; Lande & Arnold, 1983; Phillips &
83 Arnold, 1989). For example, body length may influence competitive ability or persistence,
84 whereas leg length may influence movement or the ability to remain attached to a female.

85 Directional gradients describe ~~the~~ average ~~slopes~~ slopes of the phenotype–fitness relationship,
86 while quadratic and correlational gradients describe ~~the~~ curvature and dependence between traits.
87 These latter components may reveal combinations of morphology associated with high fitness
88 even when the directional gradients for individual traits appear similar between groups.

89 Assortative mating provides a related but distinct perspective on how morphology structures
90 mating interactions. Positive assortment occurs when individuals with similar trait values pair
91 more often than expected, whereas negative assortment occurs when dissimilar individuals pair
92 preferentially (Crespi, 1989; Jiang et al., 2013). ~~Assortment~~ Assortment may arise through mutual
93 mate choice, one-sex choice, competition, spatial or temporal covariation, and mechanical
94 compatibility. It can influence the covariance between male and female phenotypes, alter the
95 distribution of mating success, and thereby affect the opportunity for sexual selection and the
96 evolutionary dynamics of sexually dimorphic traits (McDonald & Pizzari, 2016). ~~An~~ The absence
97 of assortment does not imply an absence of selection on morphology: a trait may affect whether
98 an individual is paired without determining its partner’s phenotype.

99 Weevils (Curculionoidea) encompass substantial ecological and mating-system diversity across
100 approximately 62,000 species (Haran et al., 2023), including around 1,500 species endemic to
101 New Zealand (May, 1993; see also Brown, 2017). *Lyperobates* are flightless, ground-dwelling
102 New Zealand weevils typically associated with alpine and subalpine habitats (Figure 1). ~~The~~
103 study population was nocturnally active;). This poorly known taxon offers an informative field

system because paired and unpaired adults of both sexes can be sampled under the same
conditions, while flightlessness makes body and leg morphology plausible contributors to mate
searching and attachment. The study population was nocturnally active, individuals were
commonly found on kawakawa (*Piper excelsum*), and males were sometimes observed mounted
on females for extended periods. ~~I did not observe~~No systematic observations were collected on
mating duration, remating frequency, pair formation, ~~systematic female rejection or detachment~~
~~behaviour, or~~ male–male contests, ~~or female choice; consequently, the~~. The apparent absence of
resource defence is therefore consistent with, but does not demonstrate, a scramble-competitive
mating system. Here, I examine SSD and contemporary phenotypic selection in an undescribed
species of *Lyperobates* (Coleoptera: Curculionidae: Entiminae; hereafter *Lyperobates* sp. A).

I addressed four questions. First, is female body length positively associated with egg load, as
expected under fecundity selection? Second, how are body length, hind leg length, and their
combination associated with the probability that females and males were observed paired, and is
there evidence that the correlational association differs between the sexes? Third, are females
and males dimorphic in body and hind leg length, and do the contemporary selection gradients
align with the direction and magnitude of that dimorphism? Finally, is body length assortative
within observed mating pairs?

Methods

Study species and field sampling

I studied a population of an undescribed *Lyperobates* weevil (hereafter *Lyperobates* sp. ~~A~~A)
along a 20-m sampling transect bordering forest habitat on Te Pākeka/Maud Island, New
Zealand-(41.023594°S, 173.896832°E). This taxon is ~~currently~~ under formal description but can

126 be distinguished from described congeners by consistent differences in body length, colouration,
127 and elytral morphology (C.D. Kelly, personal observation). All individuals used in this study
128 conformed to the same diagnostic phenotype and are treated as a single operational taxonomic
129 unit.

130 ~~Adult weevils were~~I collected adult weevils on 11 nights between 20 April and 5 May 2007.
131 ~~Individuals were located, locating individuals~~ opportunistically by searching the ground and low
132 vegetation (≤ 2 m) along ~~a 20-m section of track bordering forest habitat~~ the transect. Sampling
133 began approximately 1 h after sunset (~~approximately about~~ 18:30 h) and continued for 2 h.
134 ~~Weevils were~~I found weevils almost exclusively on kawakawa leaves. I collected each observed
135 male–female pair ($N = 25$), together with singleton males ($N = 75$) and females ($N = 52$),
136 yielding 100 males and 77 females ~~in total~~. Because collection was opportunistic and paired
137 individuals may differ from singletons in detectability, this sample ratio should ~~not~~ be interpreted
138 cautiously and not as an estimate of the population or operational sex ratio. ~~Each~~I placed each
139 individual ~~was placed~~ in a separate 50-mL Falcon tube, assigned it a unique identification code,
140 transported ~~to ait to the New Zealand Department of Conservation~~ field laboratory on Maud
141 Island, and subsequently euthanized it by freezing.

142 ~~All individuals were~~I photographed all individuals alongside a ruler for scale. ~~Females~~I dissected
143 females ($N = 69$) ~~were dissected~~, and ~~egg load was~~ quantified egg load as the number of eggs
144 present in the reproductive tract at the time of collection. Egg load provides an index of current
145 reproductive investment, but it may be influenced by female age, mating history, oviposition
146 history, and the timing of egg maturation; it is therefore not equivalent to lifetime realized
147 fecundity. ~~Morphological measurements were obtained from digital images~~I measured
148 morphology digitally from the scaled photographs using the segmented-line tool in Fiji (version

2.16.0, Schindelin et al., 2012). ~~Body length was~~, rather than with callipers. I measured body length from the anterior edge of the thorax to the posterior edge of the abdomen, and hind leg length ~~was measured~~ on the third leg pair from the proximal end of the femur to the distal end of the tibia, to the nearest 0.01 mm.

Ethical note

This study involved ~~behavioural observations~~field collection and morphological ~~measurements~~measurement of adult *Lyperobates* sp. A weevils. ~~Efforts to minimize stress and disturbance included minimizing handling and maintaining;~~ individuals ~~under appropriate conditions during the study. Following completion of the behavioural observations, were not~~ subjected to experimental behavioural manipulation. I minimized handling and housed weevils ~~were humanely~~individually after collection to prevent unwanted interactions. I euthanized individuals by freezing before ~~dissection and~~ photography and dissection. Formal institutional animal ethics approval was not required because the study involved invertebrates. I conducted collection and handling under permit from the New Zealand Department of Conservation. All procedures complied with applicable New Zealand regulations ~~governing the collection and handling of wildlife in New Zealand~~ and ~~followed~~ the Association for the Study of Animal Behaviour/Animal Behavior Society Guidelines for the Treatment of Animals in Behavioural Research and Teaching (“Guidelines for the Ethical Treatment of Nonhuman Animals in Behavioural Research and Teaching,” 2023).

Statistical analyses

All analyses were conducted in R (R Core Team, 2025). I ~~used~~use observed pairing status to describe whether an individual was collected as part of a male–female pair (paired = 1) or as a singleton (unpaired = 0). This cross-sectional measure is a component or proxy of mating

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172 success rather than a measure of lifetime mating or reproductive success. An association with
 173 pairing status could arise through differences in mate encounter, acceptance, mounting success,
 174 pair duration, or detectability. The selection analyses therefore assume that phenotypic
 175 differences between paired and unpaired individuals contain information about their relative
 176 propensity to be observed paired during the sampling period.

177 ~~Sex~~I assessed sex differences in total body length and total hind leg length ~~were assessed~~ using
 178 separate linear models of the form `trait ~ sex`. I also used a multivariate analysis of variance,
 179 with body and hind-leg length as the joint response and sex as the predictor, to test for overall
 180 sexual dimorphism ~~in~~ across the two traits. To describe static allometry between the traits, I fitted
 181 a log-log linear model of hind leg length as a function of body length, sex, and their interaction.
 182 The body length $\times$ sex interaction tested whether the allometric slope differed between females
 183 and males.

184 ~~Phenotype~~I analysed phenotypic selection through observed pairing status ~~was analysed~~
 185 separately for males and females. The male analysis included 100 individuals, of which 25 were
 186 paired, and 75 were unpaired; the female analysis included 77 individuals, of which 25 were
 187 paired, and 52 were unpaired. For each sex, I fitted a generalized additive model using
 188 `mgcv::gam()` with the formula

189 `mated ~ s(total_body, bs = "tp", k = 10) + s(total_leg, bs = "tp", k = 10).`

190 ~~Pairing status was~~I modelled pairing status using a binomial error distribution and logit link.
 191 ~~Body and hind leg length were represented by separate~~Separate penalized thin-plate regression
 192 splines, each with a basis dimension of ($k=10$), ~~and~~represented body and hind leg length, and I
 193 selected smoothing parameters ~~were selected~~ using the `method = "GCV.Cp"` option (Wood,

2017). The models were additive on the logit scale and did not include an explicit tensor-product interaction between the two traits.

I estimated standardized selection differentials and gradients following Lande and Arnold (1983) using the *gsg* package (Morrissey & Sakrejda, 2013). ~~The GAMs were~~ fitted the GAMs to traits measured in millimetres, but *gsg* standardized the resulting differentials and gradients to units of one within-sex phenotypic standard deviation. ~~Standardized~~ I calculated standardized directional and quadratic selection differentials ~~were calculated~~ separately for each trait using `moments.differentials()` with 10,000 bootstrap replicates. Directional differentials (*S*) describe the total association between a trait and pairing status, whereas quadratic differentials (*C*) describe changes in trait variance associated with pairing status.

~~Standardized~~ I calculated standardized linear, diagonal quadratic, and correlational selection gradients ~~were calculated~~ from each fitted GAM using `gam.gradients()`. Linear gradients (β) describe the average directional association between a trait and relative pairing status after accounting for the other trait. Diagonal quadratic gradients (γ_{ii}) describe average response-scale curvature, whereas the correlational gradient (γ_{ij}) is the response-scale cross-derivative describing how the association between one trait and pairing status varies with the other trait (Brodie et al., 1995; Phillips & Arnold, 1989). Because the models used a nonlinear logit link, the response-scale correlational gradient can be nonzero even though the GAM predictor contained separate additive smooths rather than an explicit interaction term.

The `gam.gradients()` function derives gradients from the fitted absolute-fitness function, divides them by mean fitness, and standardizes them by the observed phenotypic standard deviations. The returned estimates are therefore expressed on the standardized relative-fitness

scale (Morrissey & Sakrejda, 2013). Diagonal report diagonal quadratic gradients and their standard errors ~~were reported~~ exactly as returned by `gam.gradients()`, applying no additional doubling ~~was applied~~. Standard I obtained standard errors and two-sided *P*-values ~~were obtained~~ from 10,000 parametric-bootstrap replicates (`se.method = "boot.para"`) ~~During bootstrapping,~~ fixing smoothing parameters ~~were fixed~~ during bootstrapping at the values estimated from the original data (`refit.smooth = FALSE`). Values I report values returned as zero ~~were reported~~ as $P < 0.001$. The I estimated the gradients ~~were estimated~~ separately by sex, so statistical support for a gradient in one sex but not the other does not by itself demonstrate that the corresponding gradients differed between the sexes.

To ~~evaluate whether~~ test the apparent sex contrast in correlational selection ~~was supported by a formal sex comparison directly~~, I fitted an additional pooled quadratic logistic regression. ~~Body length, standardizing body and hind leg length were standardized separately within each sex to means of zero and standard deviations of one. Pairing status was modelled as a binary response variable with a binomial error distribution and a logit link.~~ The model included sex; the linear, squared, and cross-product terms for standardized body and hind leg length both traits; and interactions between sex and each morphological term. The $\text{sex} \times \text{body length} \times \text{hind leg length}$ term tested whether the logit-scale body length $\times$ hind leg length cross-product association differed between females and males. Because the male and female in ~~an~~ each observed pair shared the same pairing event, I calculated cluster-robust standard errors with ~~observed~~ pair identity as the clustering variable; assigning unpaired individuals ~~were assigned to~~ unique singleton clusters. ~~Inference was based on a two-sided cluster-robust Wald test. This pooled test was used to assess sex heterogeneity in the cross-product association and~~ test is complementary to, but not numerically identical to, the response-scale correlational gradients returned by

gam.gradients()). ~~The I calculated the pooled model was fitted using glm, and and its cluster-~~
~~robust Wald test in R from the clustered covariance matrix and Wald test were calculated using~~
~~the sandwich (Zeileis et al., 2020) and lmtest (Zeileis & Hothorn, 2002) packages,~~
~~respectively binomial score contributions of each pair or singleton cluster.~~

For the one-dimensional visualizations, I used fitness.landscape() to calculate predicted
mean pairing probability across 25 values of the focal trait spanning one phenotypic standard
deviation below to one ~~phenotypic~~ standard deviation above its sex-specific mean. When body
length was the focal trait, I retained hind leg length ~~was retained~~ at its observed ~~values for each~~
individual values, and vice versa; ~~predictions were~~, then averaged ~~over predictions across~~ the
observed distribution of the alternate trait. Thus, these curves are marginal fitness landscapes
rather than predictions conditional on a single fixed value of the alternate trait. Shaded regions
and dashed lines show ~~the~~ central 95% parametric-bootstrap intervals based on 1,000 replicates,
with smoothing parameters held fixed.

~~The I generated the~~ two-dimensional male response surface ~~was generated~~ directly from the
fitted male GAM. ~~Predicted, calculating predicted~~ pairing probabilities ~~were calculated~~ on a 300
× 300 grid spanning the observed ranges of male body and hind leg length. ~~Predictions To avoid~~
~~displaying unsupported extrapolations, I excluded predictions~~ for trait combinations outside the
convex hull of the observed male phenotypes ~~were excluded to avoid displaying unsupported~~
~~extrapolations,~~ and I superimposed the observed male phenotypes ~~were superimposed~~ on the
surface.

~~The I analysed the~~ relationship between female body length and egg load ~~was analysed~~ using a
negative-binomial generalized linear model fitted with MASS::glm.nb() to accommodate

261 overdispersion in egg counts (Venables & Ripley, 2002). The model formula was `eggs ~`
262 `log(total_body)`, with the default logarithmic link. This analysis estimates the association
263 between body length and egg load at collection; I interpret a positive association as consistent
264 with, rather than definitive evidence of, fecundity selection.

265 I assessed body-length assortment among the 25 observed pairs by regressing standardized
266 female body length on standardized male body length (Crespi, 1989; Jiang et al., 2013). I also
267 conducted a two-sided randomization test with 10,000 iterations. In each iteration, ~~females were~~
268 randomly reassigned females among the sampled males, recalculated the regression slope ~~was~~
269 ~~recalculated~~, and compared the absolute randomized slope ~~was compared~~ with the absolute
270 observed slope. The randomization *P*-value was the proportion of iterations producing a slope at
271 least as extreme as the observed value.

272 Results

273 Sexual dimorphism

274 The sampled adults comprised 100 males and 77 females (56.5% male), but this opportunistic
275 sample cannot be used to infer the population sex ratio. Females were larger than males in both
276 total body length and total hind leg length (Figure 2). Mean body length was 9.95 ± 0.07 mm (n
277 $= 77$) in females and 8.95 ± 0.04 mm ($n = 100$) in males, so females were approximately 11.2%
278 longer (female:male ratio = 1.11). Males were, on average, 1.00 mm shorter than females ($\beta = -$
279 1.00 ± 0.08 , $t = -13.34$, $p < 0.001$; $R^2 = 0.50$).

280 Females also had longer hind legs than males (7.05 ± 0.04 mm versus 6.68 ± 0.03 mm), a
281 difference of 5.5% (female:male ratio = 1.06). This difference was statistically supported ($\beta = -$
282 0.37 ± 0.05 , $t = -6.95$, $p < 0.001$; $R^2 = 0.22$). The multivariate sex effect was also supported

(MANOVA: Pillai's trace = 0.57, $F_{2,174} = 117.46$, $p < 0.001$). Hind leg length increased with body length in the static allometric model, but the log-log slope did not differ clearly between females and males (body length $\times$ sex: $\beta = 0.10 \pm 0.08$, $t = 1.22$, $p = 0.223$).

Female body length and egg load

Female egg load increased with body length (Figure 2Figure 3). Egg number was positively associated with log-transformed body length ($\beta = 4.50 \pm 1.33$, $z = 3.39$, $p < 0.001$). This association is consistent with fecundity selection favouring greater female body length, while recognizing that although egg load is a snapshot of current reproductive state rather than lifetime fecundity.

Phenotypic selection through observed pairing status

The directional selection gradients on body and hind leg length through observed pairing status were qualitatively similar in females and males (Table 1; Figure 3Figure 4). In both sexes, body length was positively associated with pairing status (males: $\beta_{\text{body}} = 0.97 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{body}} = 0.93 \pm 0.39$, $p = 0.032$), whereas hind leg length was negatively associated with pairing status (males: $\beta_{\text{leg}} = -0.95 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{leg}} = -1.22 \pm 0.33$, $p < 0.001$). The two sexes therefore did not experience opposing directional selection regimes on these traits. These associations may reflect differences in encounter, mounting, attachment duration, or otheranother unmeasured processesprocess and should not be interpreted automatically interpreted as differences in lifetime reproductive success.

Despite the qualitative similarity of the directional gradients, the male analysis also supported nonlinear and correlational selection (Table 1; Figure 4)(Table 1; Figure 5). Both male traits had significant positive diagonal quadratic gradients (males: $\gamma_{\text{ii_body}} = 0.60 \pm 0.35$, $p < 0.001$; $\gamma_{\text{ii_leg}} = 0.58 \pm 0.36$, $p < 0.001$) and there was significant negative correlational selection between male

body and hind leg length ($\gamma_{ij} = -0.59 \pm 0.34$, $p < 0.001$). The fitted surface therefore indicated that, on the probability scale, the association between either male trait and observed pairing status varied with the value of the other trait. Within the observed phenotypic range, the predicted pairing probability was highest among males ~~that combined~~ combining relatively large body length with short hind legs.

The female analysis provided no statistical support for nonlinear selection on body length ($\gamma_{ii_body} = -0.59 \pm 2.16$, $p = 0.391$), nonlinear selection on hind leg length ($\gamma_{ii_leg} = 0.45 \pm 0.41$, $p = 0.105$), or correlational selection between the traits ($\gamma_{ij} = -0.23 \pm 0.48$, $p = 0.414$). ~~However, the~~ The difference in statistical support between the separate ~~male and female~~ sex-specific analyses did not establish a sex difference. In the pooled quadratic ~~logistic~~ model, the sex $\times$ body length $\times$ hind leg length term provided little evidence that the logit-scale cross-product association differed between ~~the sexes~~ females and males ($z = -1.31$, $P = 0.190$). Thus, the correlational pattern was supported within males, but the data did not clearly demonstrate that it was stronger in males than in females.

~~Sexual dimorphism~~

~~The sampled adults comprised 100 males and 77 females (56.5% male), but this opportunistic sample cannot be used to infer the population sex ratio. Females were larger than males in both total body length and total hind leg length (Figure 5). Mean body length was 9.95 ± 0.07 mm ($n = 77$) in females and 8.95 ± 0.04 mm ($n = 100$) in males, so females were approximately 11.2% longer (female:male ratio = 1.11). Males were, on average, 1.00 mm shorter than females ($\beta = 1.00 \pm 0.08$, $t = 13.34$, $p < 0.001$; $R^2 = 0.50$).~~

~~Females also had longer hind legs than males (7.05 ± 0.04 mm versus 6.68 ± 0.03 mm), a difference of 5.5% (female:male ratio = 1.06). This difference was statistically supported ($\beta =$~~

~~0.37 ± 0.05, $t = -6.95$, $p < 0.001$; $R^2 = 0.22$). The multivariate sex effect was also supported (MANOVA: Pillai's trace = 0.57, $F_{2,174} = 117.46$, $p < 0.001$).~~

Assortative mating

There was no evidence of assortative mating ~~in~~for body length ~~among~~within the 25 observed pairs. The relationship between male and female body length was weak and non-significant ($\beta = -0.23 \pm 0.20$, $t = -1.14$, $p = 0.27265$), and the observed slope did not differ from the randomized distribution ($P = 0.27271$). Thus, body length did not measurably structure which sampled male was paired with which sampled female.

Discussion

Females were larger than males in both body and hind leg length, and ~~female body length longer~~ females carried more eggs. Observed pairing was ~~positively associated also nonrandom~~ with egg load. ~~Based on observed respect to morphology in both sexes:~~ pairing status, ~~the directional selection gradients on~~ increased with body length and decreased with hind leg length ~~were qualitatively similar in females and males: body length was positively associated with pairing, and hind leg length was negatively associated with pairing in both sexes. The two sexes therefore did not experience opposing directional selection regimes on these traits.~~ The male analysis additionally supported nonlinear and correlational selection, with the highest predicted pairing probability among ~~males being for combinations of~~ relatively large ~~bodies and bodied~~, short ~~hind legs, whereas the corresponding female gradients were not statistically supported. Nevertheless, the legged males, although the~~ pooled comparison ~~provided little evidence that the body length × hind leg length~~ did not clearly demonstrate that this association differed between the sexes. The ~~apparent contrast between the separate analyses therefore should not be interpreted as~~

~~demonstrating male-specific correlational selection.~~ Body length was not assortative within observed pairs. ~~My findings document female-biased SSD and several contemporary morphology–fitness component associations, but they do not demonstrate that the measured selection components caused or currently maintain the dimorphism.~~ Together, the results identify concordant directional pairing-status selection in females and males, a fecundity-related advantage of female size, and a complex multivariate pairing surface.

The positive association between female body length and egg load is consistent with a common mechanism proposed for female-biased SSD: larger females can allocate more space or resources to egg production (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017). The inference is nevertheless narrower than a direct estimate of lifetime fecundity selection. The number of eggs present at dissection represents current egg load and may vary with age, reproductive stage, previous oviposition, and the rate at which eggs mature (e.g. Kelly et al., 2024). Consequently, the ~~result shows~~ results show that longer females carried more eggs at collection; ~~it does~~ they do not establish how body length affects lifetime egg production, offspring recruitment, or population growth. Repeated measures of oviposition or lifetime reproductive output would provide a stronger test.

~~The associations between morphology and~~ Several behavioural mechanisms could generate these pairing patterns. Larger paired males could reflect female preference or greater male success in competition, mounting, or remaining attached. Larger paired females could reflect male choice because female body length predicted egg load, although size-dependent encounter rates or pair duration could generate the same association. Selection through observed pairing status ~~also require careful interpretation. Paired males were not followed across a mating season, and the survey did not record the number of females encountered, copulations completed, offspring~~

~~sired, or the duration for which each male had been mounted,~~is therefore consistent with female choice for larger males, male choice for larger females, or mutual choice, but does not distinguish among them. The negative hind-leg gradients and male response surface additionally suggest testable hypotheses involving movement, leverage, stability while mounted, or attachment endurance.

Pair duration and sexual conflict are especially relevant alternatives. In the seed beetle *Callosobruchus maculatus*, spined male genitalia damage the female reproductive tract and female kicking shortens copulation and reduces damage (Crudgington & Siva-Jothy, 2000). I did not quantify attachment duration in *Lyperobates* sp. A or collect systematic observations of kicking, induced detachment, genital morphology, or reproductive-tract scarring. A phenotype associated with prolonged attachment could therefore be overrepresented among observed pairs even ~~if it did not increase~~without increasing the number of matings or fertilizations. ~~The male gradients may therefore reflect mate searching, mounting, female acceptance, attachment duration, detectability, or some combination~~Direct observations of these processes. Likewise, the field observations are consistent with a scramble-like system because males were not seen defending resources, and pair formation ~~and male-male interactions were not systematically observed~~. The present data cannot exclude female choice, interference among males, or other ~~forms of competition~~, detachment, remating, and copulation duration would distinguish mate acquisition from persistence and sexual conflict.

The qualitative similarity of the directional gradients between the sexes is ~~itself~~ a central feature of ~~these~~the results. Both females and males showed positive body-length gradients and negative hind-leg gradients ~~through pairing status~~, with similar estimated magnitudes. ~~The~~ Although the male analysis supported ~~male~~a correlational gradient ~~adds a second layer: within males, the~~

association between body length and pairing depended on hind leg length, and the highest predicted pairing probabilities occurred among relatively large-bodied, short-legged males. The biological mechanism behind this combination is unresolved, and several hypotheses—concerning movement, leverage, stability while mounted, or attachment duration—are plausible but not testable with these data. Although the corresponding correlational female gradient was not supported in females, the pooled sex comparison provided little evidence that the body length $\times$ hind leg length cross-product association differed between females and males. The apparent male-only statistical support result may therefore reflect sampling variation or lower precision in the female analysis rather than a genuinely sex-specific selection surface. At the same time, failure to detect an interaction is not evidence that the sexes are equivalent, and the uncertainty around the sex contrast leaves room for biologically meaningful differences. Existing dimorphism may therefore could instead reflect past selection acting in earlier generations, sex-specific viability selection, development, ecological differences ecology, genetic architecture, or fitness components not measured here (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007). Inbreeding is another potential modifier: it can depress male morphology and reproduction (Vasudeva et al., 2025), but relatedness and genetic load were not measured in this population and cannot be inferred from the restricted spatial sample. The greater proportional dimorphism in body length than in hind leg length also likewise does not map straightforwardly onto the contemporary gradients, and there was no clear sex difference in the static allometric slope relating the two traits. This mismatch is informative because it cautions against inferring the evolutionary illustrates why a single fitness component cannot by itself identify the historical cause of SSD from a single episode or component of phenotypic selection.

419 The absence of body-length assortment indicates that, among the 25 pairs sampled, large males
420 were no more likely to be paired with large females than expected by chance. This result does
421 not contradict selection through pairing status. Morphology may affect the probability of being
422 paired while having little effect on the phenotype of the eventual partner (McDonald & Pizzari,
423 2016). The modest number of pairs nevertheless limits power to detect weak assortment, and a
424 single nocturnal snapshot cannot reveal whether assortment varies through the season or across
425 sites.

426 ~~The scope of inference main inferential limit is necessarily limited to this population, sampling
427 interval, and taxon. A single species result cannot establish a general rule for flightless weevils
428 or scramble-competitive arthropods. Its broader contribution is instead comparative: it adds an
429 example in which female-biased SSD co-occurs with a positive body-length-egg-load
430 association, similar directional pairing-status gradients in the sexes, and a male analysis that
431 supported a complex multivariate surface without clear evidence from the pooled comparison
432 that this surface differed between the sexes. Determining whether these features recur in related
433 weevils will require replicated sampling across populations, dates, and species, ideally combined
434 with direct observation of mate searching, pair formation, pair duration, and reproductive
435 success.~~

436 ~~Several limitations are therefore central rather than peripheral. The 25 observed pairs came from
437 one site and a restricted 11-night period; pairing status was binary and cross-sectional; design: 25
438 pairs were sampled at one site during 11 nights, egg load was measured once, and pairing
439 behaviour linking morphology to pairing was not recorded directly. The opportunistically
440 collected Pairing status is consequently a measure of observed precopulatory success, not lifetime
441 mating or reproductive success, and the sampled sex ratio may also reflect ~~sex-specific~~ activity~~

or detectability rather than ~~population~~-abundance. ~~These constraints increase uncertainty about~~
~~temporal repeatability and the biological processes underlying the statistical associations. Future~~
~~work should combine repeated~~The study nevertheless provides comparative field surveys with
~~behavioural estimates for both sexes of a poorly known species. Replication across dates,~~
~~populations, and related weevils, combined with~~ observations, ~~experimental measures of~~
~~locomotion and~~ or experiments on mate searching, choice, attachment, and ~~more complete~~
~~estimates of reproductive success for females and males~~output, would test the mechanisms
~~suggested by the present patterns.~~

In conclusion, ~~this study documents female-biased SSD, a positive association between female~~
~~body length and egg load, and qualitatively similar directional gradients in body and hind-leg~~
~~length, as observed in the pairing status of female and male *Lyperobates* sp. A-A paired~~
~~nonrandomly with respect to size: longer-bodied, shorter-legged individuals of both sexes were~~
~~more likely to be observed paired. Larger females also carried more eggs, and females were~~
~~larger than males in both traits.~~ The male analysis ~~supported a dependence of pairing probability~~
~~on the combination of body and hind-leg length, whereas the corresponding female gradient was~~
~~not statistically supported. However,~~ further identified a high-pairing region combining large
~~bodies with short legs, although~~ the pooled comparison ~~provided little evidence~~did not establish
that ~~this~~the correlational association differed between the sexes, ~~so the result does not establish~~
male-specific correlational selection. The cross-sectional measures of egg load and pairing
status, together with the qualitative similarity of the directional gradients between the sexes, also
prevent a strong causal inference that these-. Size therefore matters for both female reproductive
investment and pairing in this wild population, while the correspondence between contemporary
associations produced the observed SSD. The results are best interpreted as evidence that

465 ~~contemporary components of selection in this population are incompletely aligned with~~selection
466 and existing ~~sexual~~female-biased dimorphism remains incomplete.
467

Data availability

Data are available through the OSF repository at

https://osf.io/khgg5/overview?view_only=728733529d06410ab17086f648a49a38.

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Table 1: Selection differentials (S, C) and standardized linear (β) and diagonal quadratic (γ) selection gradients through observed pairing status for female and male *Lyperobates* sp. A weevils. Estimates are means $\pm$ SE. Differentials quantify total phenotypic selection, whereas gradients describe associations with each trait after accounting for the other trait. Estimates were returned directly by `gsg::gam.gradients()` on the relative-fitness scale; diagonal quadratic gradients and SEs were not doubled. Standard errors and two-sided *P*-values were obtained from 10,000 parametric-bootstrap replicates.

Trait	Differentials				Gradients			
	$S_i \pm SE$	<i>P</i>	$C_{ii} \pm SE$	<i>P</i>	$\beta_i \pm SE$	<i>P</i>	$\gamma \pm SE$	<i>P</i>
(a) Females								
body length	-0.040 $\pm$ 0.146	0.788	-0.388 $\pm$ 0.220	0.090	0.930 $\pm$ 0.391	0.032	-0.593 $\pm$ 2.163	0.391
hind leg length	-0.148 $\pm$ 0.153	0.335	-0.246 $\pm$ 0.217	0.283	-1.224 $\pm$ 0.332	<0.001	0.454 $\pm$ 0.406	0.105
body length $\times$ hind leg length							-0.228 $\pm$ 0.483	0.414
(b) Males								
body length	0.210 $\pm$ 0.165	0.205	-0.092 $\pm$ 0.199	0.620	0.968 $\pm$ 0.247	<0.001	0.601 $\pm$ 0.354	<0.001
hind leg length	-0.175 $\pm$ 0.165	0.289	-0.121 $\pm$ 0.213	0.553	-0.954 $\pm$ 0.251	<0.001	0.584 $\pm$ 0.360	<0.001
body length $\times$ hind leg length							-0.593 $\pm$ 0.335	<0.001

Figure captions

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Figure 1: Mating pair of *Lyperobates* sp. A weevils. The male is mounted dorsally on the female in a typical copulatory position. (a) Lateral (side) view showing the male positioned atop the female's abdomen, with legs grasping the female's body. (b) Frontal view illustrating the alignment of the pair and relative positioning of the male on the female's dorsum.

Figure 2: Relationship between female body length and egg load. Points represent individual females, and egg load is the number of eggs present in the reproductive tract at dissection. Body length is plotted on a log scale. The line shows the fitted mean from the negative-binomial generalized linear model, and the shaded region is the 95% confidence interval. Egg load increased with body length; this cross-sectional measure is not equivalent to lifetime realized fecundity.

Figure 3: Associations between morphology and observed pairing status in males (a–b) and females (c–d). Panels show the predicted probability of being observed paired from binomial generalized additive models as a function of body length (a, c) and hind leg length (b, d), with the alternate trait held constant. Solid lines are model predictions and dashed lines are 95% intervals estimated by parametric bootstrapping. In both sexes, body length had a positive directional gradient and hind leg length had a negative directional gradient. Pairing status is a cross-sectional proxy and may reflect encounter, mounting, or pair duration rather than lifetime reproductive success.

Figure 4: Predicted probability that a male was observed paired as a function of body length and leg length. Colours and labelled contours represent fitted probabilities from the generalized additive model. Points show observed males, and predictions are restricted to the convex hull of the observed combinations of body and leg length. Higher pairing probabilities were associated with greater body length relative to leg length.

Figure 5: Sexual dimorphism in body and hind leg length. Boxplots show total body length and total hind leg length (mm) for females (light grey) and males (dark grey), with points representing individuals. Horizontal lines are medians, boxes are interquartile ranges, and whiskers extend to 1.5× the interquartile range. Females were significantly larger than males for both traits, as assessed by the linear models reported in the Results.

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Figure 4: Associations between morphology and observed pairing status in males (a–b) and females (c–d). Panels show the predicted probability of being observed paired from binomial generalized additive models as a function of body length (a, c) and hind leg length (b, d), averaged over the observed distribution of the alternate trait. Solid lines are model predictions and dashed lines are 95% intervals estimated by parametric bootstrapping. In both sexes, body length had a positive directional gradient and hind leg length had a negative directional gradient. Pairing status is a cross-sectional proxy and may reflect encounter, mounting, or pair duration rather than lifetime reproductive success.

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Figure 5: Predicted probability that a male was observed paired as a function of body length and leg length. Colours and labelled contours represent fitted probabilities from the generalized additive model. White circles show unpaired males and yellow triangles show paired males; predictions are restricted to the convex hull of the observed combinations of body and leg length. Higher pairing probabilities were associated with greater body length relative to leg length.

1 **Sexual size dimorphism and multivariate phenotypic**
2 **selection in a flightless weevil**

3

Abstract

Body size can affect fecundity and access to mates in both sexes, yet field estimates of these selection components are uncommon for females and males in the same population. I sampled paired and unpaired adults of an undescribed flightless New Zealand weevil (*Lyperobates* sp. A) and examined body length, hind leg length, pairing status, female egg load, sexual dimorphism, and size assortment. Females were larger than males in both morphological traits, and longer females carried more eggs, consistent with fecundity selection favouring greater female size. In both sexes, body length was positively associated and hind leg length negatively associated with observed pairing status. The male analysis also supported nonlinear and correlational selection, with the highest predicted pairing probability among relatively large-bodied, short-legged males. A pooled comparison, however, did not clearly demonstrate that this correlational association differed between the sexes. Body length was not assortative within the 25 observed pairs. These results reveal size-dependent nonrandom pairing in both sexes and a fecundity-related advantage of female body size. They show that contemporary components of selection may contribute to female-biased dimorphism without mapping simply onto the magnitude of the observed sex differences.

Introduction

Sexual size dimorphism (SSD) is a widespread and evolutionarily significant feature of animal morphology (Darwin, 1871; Fairbairn et al., 2007). Its direction and magnitude can reflect sex-specific fecundity and sexual selection, as well as viability selection, ecological divergence, developmental processes, and genetic constraints or correlations between the sexes (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007). Female-biased SSD is particularly common among insects and is often associated with fecundity selection because larger females can produce or carry more eggs (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017; Zhu et al., 2025). A positive size–fecundity relationship, however, does not by itself demonstrate that fecundity selection produced SSD. Dimorphism is an evolved outcome of selection acting across generations, whereas estimates from one population and sampling period describe contemporary associations with particular components of fitness.

Sexual selection on body size can also operate in either sex. When female fecundity increases with size, males may benefit by preferentially courting, mounting, or investing in larger females (Bonduriansky, 2001). In the curculionid beetle *Diaprepes abbreviatus*, for example, males preferred larger, more fecund females, females accepted large males more readily, and competition contributed to size-assortative mating (Harari et al., 1999). Nonrandom pairing by size may therefore reflect female choice, male choice, mutual choice, competition, or some combination of these processes.

The sexes nevertheless share most of their genome, which can constrain independent responses to sex-specific selection. Experimental work on the seed beetle *Callosobruchus maculatus* revealed a strong intersexual autosomal genetic correlation for body size, while sex-linked

variance enabled SSD to respond to selection (Kaufmann et al., 2021). Inbreeding can further alter phenotype–fitness relationships by exposing deleterious recessive variants; for example, six generations of full-sib mating in *Tribolium castaneum* reduced male body size, testis size, sperm length, lifespan, and reproductive output (Vasudeva et al., 2025). Population structure, relatedness, and genetic load could consequently influence sex-specific morphology and fitness, although evaluating those mechanisms requires pedigree or genomic information.

When sexual selection favours increased male body length, as often occurs under male–male contest competition, male-biased SSD may evolve, or female-biased SSD may be reduced (Andersson, 1994; Emlen & Oring, 1977). Conversely, female-biased SSD has sometimes been attributed to weak sexual selection on males or to selection favouring smaller, more mobile males (Fairbairn et al., 2007; Ghiselin, 1974). Scramble-competition polygyny is frequently invoked in this context because male reproductive success can depend on locating and securing females rather than defending resources (Andersson, 1994; Moya-Laraño et al., 2002). Smaller males, or males with locomotor morphology that facilitates mate searching, have higher mating success in some systems, including Japanese beetles (*Popillia japonica*) and Cook Strait giant weta (*Deinacrida rugosa*) (Kelly et al., 2008; Kelly, 2020; Kelly & Gwynne, 2023).

Comparative evidence, however, shows that larger males are also favoured in many scramble-competitive arthropods (Herberstein et al., 2017). The direction of selection on male morphology therefore cannot be inferred from a broad mating-system label alone, especially when mate searching and pair formation have not been observed directly.

Multivariate analyses are important because the association between one trait and a fitness component can depend on other traits (Brodie et al., 1995; Lande & Arnold, 1983; Phillips & Arnold, 1989). For example, body length may influence competitive ability or persistence,

67 whereas leg length may influence movement or the ability to remain attached to a female.
68 Directional gradients describe average slopes of the phenotype–fitness relationship, while
69 quadratic and correlational gradients describe curvature and dependence between traits. These
70 latter components may reveal combinations of morphology associated with high fitness even
71 when the directional gradients for individual traits appear similar between groups.

72 Assortative mating provides a related but distinct perspective on how morphology structures
73 mating interactions. Positive assortment occurs when individuals with similar trait values pair
74 more often than expected, whereas negative assortment occurs when dissimilar individuals pair
75 preferentially (Crespi, 1989; Jiang et al., 2013). Assortment may arise through mutual mate
76 choice, one-sex choice, competition, spatial or temporal covariation, and mechanical
77 compatibility. It can influence the covariance between male and female phenotypes, alter the
78 distribution of mating success, and thereby affect the opportunity for sexual selection and the
79 evolutionary dynamics of sexually dimorphic traits (McDonald & Pizzari, 2016). The absence of
80 assortment does not imply an absence of selection on morphology: a trait may affect whether an
81 individual is paired without determining its partner’s phenotype.

82 Weevils (Curculionoidea) encompass substantial ecological and mating-system diversity across
83 approximately 62,000 species (Haran et al., 2023), including around 1,500 species endemic to
84 New Zealand (May, 1993; see also Brown, 2017). *Lyperobates* are flightless, ground-dwelling
85 New Zealand weevils typically associated with alpine and subalpine habitats (Figure 1). This
86 poorly known taxon offers an informative field system because paired and unpaired adults of
87 both sexes can be sampled under the same conditions, while flightlessness makes body and leg
88 morphology plausible contributors to mate searching and attachment. The study population was
89 nocturnally active, individuals were commonly found on kawakawa (*Piper excelsum*), and males

were sometimes observed mounted on females for extended periods. No systematic observations were collected on mating duration, remating frequency, pair formation, female rejection or detachment behaviour, or male–male contests. The apparent absence of resource defence is therefore consistent with, but does not demonstrate, a scramble-competitive mating system. Here, I examine SSD and contemporary phenotypic selection in an undescribed species of *Lyperobates* (Coleoptera: Curculionidae: Entiminae; hereafter *Lyperobates* sp. A).

I addressed four questions. First, is female body length positively associated with egg load, as expected under fecundity selection? Second, how are body length, hind leg length, and their combination associated with the probability that females and males were observed paired, and is there evidence that the correlational association differs between the sexes? Third, are females and males dimorphic in body and hind leg length, and do the contemporary selection gradients align with the direction and magnitude of that dimorphism? Finally, is body length assortative within observed mating pairs?

Methods

Study species and field sampling

I studied a population of an undescribed *Lyperobates* weevil (hereafter *Lyperobates* sp. A) along a 20-m sampling transect bordering forest habitat on Te Pākeka/Maud Island, New Zealand (41.023594°S, 173.896832°E). This taxon is under formal description but can be distinguished from described congeners by consistent differences in body length, colouration, and elytral morphology (C.D. Kelly, personal observation). All individuals used in this study conformed to the same diagnostic phenotype and are treated as a single operational taxonomic unit.

111 I collected adult weevils on 11 nights between 20 April and 5 May 2007, locating individuals
112 opportunistically by searching the ground and low vegetation (≤ 2 m) along the transect.
113 Sampling began approximately 1 h after sunset (about 18:30 h) and continued for 2 h. I found
114 weevils almost exclusively on kawakawa leaves. I collected each observed male–female pair (N
115 = 25), together with singleton males ($N = 75$) and females ($N = 52$), yielding 100 males and 77
116 females. Because collection was opportunistic and paired individuals may differ from singletons
117 in detectability, this sample ratio should be interpreted cautiously and not as an estimate of the
118 population or operational sex ratio. I placed each individual in a separate 50-mL Falcon tube,
119 assigned it a unique identification code, transported it to the New Zealand Department of
120 Conservation field laboratory on Maud Island, and subsequently euthanized it by freezing.

121 I photographed all individuals alongside a ruler for scale. I dissected females ($N = 69$) and
122 quantified egg load as the number of eggs present in the reproductive tract at the time of
123 collection. Egg load provides an index of current reproductive investment, but it may be
124 influenced by female age, mating history, oviposition history, and the timing of egg maturation;
125 it is therefore not equivalent to lifetime realized fecundity. I measured morphology digitally from
126 the scaled photographs using the segmented-line tool in Fiji (version 2.16.0, Schindelin et al.,
127 2012), rather than with callipers. I measured body length from the anterior edge of the thorax to
128 the posterior edge of the abdomen, and hind leg length on the third leg pair from the proximal
129 end of the femur to the distal end of the tibia, to the nearest 0.01 mm.

130 **Ethical note**

131 This study involved field collection and morphological measurement of adult *Lyperobates* sp. A
132 weevils; individuals were not subjected to experimental behavioural manipulation. I minimized
133 handling and housed weevils individually after collection to prevent unwanted interactions. I

134 euthanized individuals by freezing before photography and dissection. Formal institutional
135 animal ethics approval was not required because the study involved invertebrates. I conducted
136 collection and handling under permit from the New Zealand Department of Conservation. All
137 procedures complied with applicable New Zealand regulations and the Association for the Study
138 of Animal Behaviour/Animal Behavior Society Guidelines for the Treatment of Animals in
139 Behavioural Research and Teaching (“Guidelines for the Ethical Treatment of Nonhuman
140 Animals in Behavioural Research and Teaching,” 2023).

141 **Statistical analyses**

142 All analyses were conducted in R (R Core Team, 2025). I use observed pairing status to describe
143 whether an individual was collected as part of a male–female pair (paired = 1) or as a singleton
144 (unpaired = 0). This cross-sectional measure is a component or proxy of mating success rather
145 than a measure of lifetime mating or reproductive success. An association with pairing status
146 could arise through differences in mate encounter, acceptance, mounting success, pair duration,
147 or detectability. The selection analyses therefore assume that phenotypic differences between
148 paired and unpaired individuals contain information about their relative propensity to be
149 observed paired during the sampling period.

150 I assessed sex differences in total body length and total hind leg length using separate linear
151 models of the form $\text{trait} \sim \text{sex}$. I also used a multivariate analysis of variance, with body and
152 hind leg length as the joint response and sex as the predictor, to test for overall sexual
153 dimorphism across the two traits. To describe static allometry between the traits, I fitted a log–
154 log linear model of hind leg length as a function of body length, sex, and their interaction. The
155 body length $\times$ sex interaction tested whether the allometric slope differed between females and
156 males.

157 I analysed phenotypic selection through observed pairing status separately for males and
 158 females. The male analysis included 100 individuals, of which 25 were paired and 75 were
 159 unpaired; the female analysis included 77 individuals, of which 25 were paired and 52 were
 160 unpaired. For each sex, I fitted a generalized additive model using `mgcv::gam()` with the
 161 formula
 162 `mated ~ s(total_body, bs = "tp", k = 10) + s(total_leg, bs = "tp", k = 10)`.
 163 I modelled pairing status using a binomial error distribution and logit link. Separate penalized
 164 thin-plate regression splines, each with a basis dimension of ($k=10$), represented body and hind
 165 leg length, and I selected smoothing parameters using the `method = "GCV.Cp"` option (Wood,
 166 2017). The models were additive on the logit scale and did not include an explicit tensor-product
 167 interaction between the two traits.
 168 I estimated standardized selection differentials and gradients following Lande and Arnold (1983)
 169 using the *gsg* package (Morrissey & Sakrejda, 2013). I fitted the GAMs to traits measured in
 170 millimetres, but *gsg* standardized the resulting differentials and gradients to units of one within-
 171 sex phenotypic standard deviation. I calculated standardized directional and quadratic selection
 172 differentials separately for each trait using `moments.differentials()` with 10,000 bootstrap
 173 replicates. Directional differentials (S) describe the total association between a trait and pairing
 174 status, whereas quadratic differentials (C) describe changes in trait variance associated with
 175 pairing status.
 176 I calculated standardized linear, diagonal quadratic, and correlational selection gradients from
 177 each fitted GAM using `gam.gradients()`. Linear gradients (β) describe the average directional
 178 association between a trait and relative pairing status after accounting for the other trait.

Diagonal quadratic gradients (γ_{ii}) describe average response-scale curvature, whereas the correlational gradient (γ_{ij}) is the response-scale cross-derivative describing how the association between one trait and pairing status varies with the other trait (Brodie et al., 1995; Phillips & Arnold, 1989). Because the models used a nonlinear logit link, the response-scale correlational gradient can be nonzero even though the GAM predictor contained separate additive smooths rather than an explicit interaction term.

The `gam.gradients()` function derives gradients from the fitted absolute-fitness function, divides them by mean fitness, and standardizes them by the observed phenotypic standard deviations. The returned estimates are therefore expressed on the standardized relative-fitness scale (Morrissey & Sakrejda, 2013). I report diagonal quadratic gradients and their standard errors exactly as returned by `gam.gradients()`, applying no additional doubling. I obtained standard errors and two-sided P -values from 10,000 parametric-bootstrap replicates (`se.method = "boot.para"`), fixing smoothing parameters during bootstrapping at the values estimated from the original data (`refit.smooth = FALSE`). I report values returned as zero as $P < 0.001$. I estimated the gradients separately by sex, so statistical support for a gradient in one sex but not the other does not by itself demonstrate that the corresponding gradients differed between the sexes.

To test the apparent sex contrast in correlational selection directly, I fitted an additional pooled quadratic logistic regression, standardizing body and hind leg length separately within each sex. The model included sex; the linear, squared, and cross-product terms for both traits; and interactions between sex and each morphological term. The $\text{sex} \times \text{body length} \times \text{hind leg length}$ term tested whether the logit-scale cross-product association differed between females and males. Because the male and female in each observed pair shared the same pairing event, I

202 calculated cluster-robust standard errors with pair identity as the clustering variable, assigning
203 unpaired individuals to unique singleton clusters. This test is complementary to, but not
204 numerically identical to, the response-scale gradients returned by `gam.gradients()`. I calculated
205 the pooled model and its cluster-robust Wald test in R from the binomial score contributions of
206 each pair or singleton cluster.

207 For the one-dimensional visualizations, I used `fitness.landscape()` to calculate predicted
208 mean pairing probability across 25 values of the focal trait spanning one phenotypic standard
209 deviation below to one standard deviation above its sex-specific mean. When body length was
210 the focal trait, I retained hind leg length at its observed individual values, and vice versa, then
211 averaged predictions across the observed distribution of the alternate trait. Thus, these curves are
212 marginal fitness landscapes rather than predictions conditional on a single fixed value of the
213 alternate trait. Shaded regions and dashed lines show central 95% parametric-bootstrap intervals
214 based on 1,000 replicates, with smoothing parameters held fixed.

215 I generated the two-dimensional male response surface directly from the fitted male GAM,
216 calculating predicted pairing probabilities on a 300×300 grid spanning the observed ranges of
217 male body and hind leg length. To avoid displaying unsupported extrapolations, I excluded
218 predictions for trait combinations outside the convex hull of the observed male phenotypes, and I
219 superimposed the observed male phenotypes on the surface.

220 I analysed the relationship between female body length and egg load using a negative-binomial
221 generalized linear model fitted with `MASS::glm.nb()` to accommodate overdispersion in egg
222 counts (Venables & Ripley, 2002). The model formula was `eggs ~ log(total_body)`, with the
223 default logarithmic link. This analysis estimates the association between body length and egg

load at collection; I interpret a positive association as consistent with, rather than definitive evidence of, fecundity selection.

I assessed body-length assortment among the 25 observed pairs by regressing standardized female body length on standardized male body length (Crespi, 1989; Jiang et al., 2013). I also conducted a two-sided randomization test with 10,000 iterations. In each iteration, I randomly reassigned females among the sampled males, recalculated the regression slope, and compared the absolute randomized slope with the absolute observed slope. The randomization *P*-value was the proportion of iterations producing a slope at least as extreme as the observed value.

Results

Sexual dimorphism

The sampled adults comprised 100 males and 77 females (56.5% male), but this opportunistic sample cannot be used to infer the population sex ratio. Females were larger than males in both total body length and total hind leg length (Figure 2). Mean body length was 9.95 ± 0.07 mm ($n = 77$) in females and 8.95 ± 0.04 mm ($n = 100$) in males, so females were approximately 11.2% longer (female:male ratio = 1.11). Males were, on average, 1.00 mm shorter than females ($\beta = -1.00 \pm 0.08$, $t = -13.34$, $p < 0.001$; $R^2 = 0.50$).

Females also had longer hind legs than males (7.05 ± 0.04 mm versus 6.68 ± 0.03 mm), a difference of 5.5% (female:male ratio = 1.06). This difference was statistically supported ($\beta = -0.37 \pm 0.05$, $t = -6.95$, $p < 0.001$; $R^2 = 0.22$). The multivariate sex effect was also supported (MANOVA: Pillai's trace = 0.57, $F_{2,174} = 117.46$, $p < 0.001$). Hind leg length increased with body length in the static allometric model, but the log-log slope did not differ clearly between females and males (body length $\times$ sex: $\beta = 0.10 \pm 0.08$, $t = 1.22$, $p = 0.223$).

Female body length and egg load

Female egg load increased with body length (Figure 3). Egg number was positively associated with log-transformed body length ($\beta = 4.50 \pm 1.33$, $z = 3.39$, $p < 0.001$). This association is consistent with fecundity selection favouring greater female body length, although egg load is a snapshot of current reproductive state rather than lifetime fecundity.

Phenotypic selection through observed pairing status

The directional selection gradients on body and hind leg length through observed pairing status were qualitatively similar in females and males (Table 1; Figure 4). In both sexes, body length was positively associated with pairing status (males: $\beta_{\text{body}} = 0.97 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{body}} = 0.93 \pm 0.39$, $p = 0.032$), whereas hind leg length was negatively associated with pairing status (males: $\beta_{\text{leg}} = -0.95 \pm 0.25$, $p < 0.001$; females: $\beta_{\text{leg}} = -1.22 \pm 0.33$, $p < 0.001$). The two sexes therefore did not experience opposing directional selection regimes on these traits. These associations may reflect differences in encounter, mounting, attachment duration, or another unmeasured process and should not be interpreted automatically as differences in lifetime reproductive success.

Despite the qualitative similarity of the directional gradients, the male analysis also supported nonlinear and correlational selection (Table 1; Figure 5). Both male traits had significant positive diagonal quadratic gradients (males: $\gamma_{\text{ii_body}} = 0.60 \pm 0.35$, $p < 0.001$; $\gamma_{\text{ii_leg}} = 0.58 \pm 0.36$, $p < 0.001$) and there was significant negative correlational selection between male body and hind leg length ($\gamma_{\text{ij}} = -0.59 \pm 0.34$, $p < 0.001$). The fitted surface therefore indicated that, on the probability scale, the association between either male trait and observed pairing status varied with the value of the other trait. Within the observed phenotypic range, predicted pairing

probability was highest among males combining relatively large body length with short hind legs.

The female analysis provided no statistical support for nonlinear selection on body length ($\gamma_{ii_body} = -0.59 \pm 2.16$, $p=0.391$), nonlinear selection on hind leg length ($\gamma_{ii_leg} = 0.45 \pm 0.41$, $p=0.105$), or correlational selection between the traits ($\gamma_{ij} = -0.23 \pm 0.48$, $p=0.414$). The difference in statistical support between the separate sex-specific analyses did not establish a sex difference. In the pooled quadratic model, the sex $\times$ body length $\times$ hind leg length term provided little evidence that the logit-scale cross-product association differed between females and males ($z = -1.31$, $P = 0.190$). Thus, the correlational pattern was supported within males, but the data did not clearly demonstrate that it was stronger in males than in females.

Assortative mating

There was no evidence of assortative mating for body length within the 25 observed pairs. The relationship between male and female body length was weak and non-significant ($\beta = -0.23 \pm 0.20$, $t = -1.14$, $p=0.265$), and the observed slope did not differ from the randomized distribution ($P = 0.271$). Thus, body length did not measurably structure which sampled male was paired with which sampled female.

Discussion

Females were larger than males in both body and hind leg length, and longer females carried more eggs. Observed pairing was also nonrandom with respect to morphology in both sexes: pairing status increased with body length and decreased with hind leg length. The male analysis additionally supported nonlinear and correlational selection, with the highest predicted pairing probability among relatively large-bodied, short-legged males, although the pooled comparison

did not clearly demonstrate that this association differed between the sexes. Body length was not assortative within observed pairs. Together, the results identify concordant directional pairing-status selection in females and males, a fecundity-related advantage of female size, and a complex multivariate pairing surface.

The positive association between female body length and egg load is consistent with a common mechanism proposed for female-biased SSD: larger females can allocate more space or resources to egg production (Blanckenhorn, 2005; Honek, 1993; Kingsolver & Pfennig, 2004; Pincheira-Donoso & Hunt, 2017). The inference is nevertheless narrower than a direct estimate of lifetime fecundity selection. The number of eggs present at dissection represents current egg load and may vary with age, reproductive stage, previous oviposition, and the rate at which eggs mature (e.g. Kelly et al., 2024). Consequently, the results show that longer females carried more eggs at collection; they do not establish how body length affects lifetime egg production, offspring recruitment, or population growth. Repeated measures of oviposition or lifetime reproductive output would provide a stronger test.

Several behavioural mechanisms could generate these pairing patterns. Larger paired males could reflect female preference or greater male success in competition, mounting, or remaining attached. Larger paired females could reflect male choice because female body length predicted egg load, although size-dependent encounter rates or pair duration could generate the same association. Selection through observed pairing status is therefore consistent with female choice for larger males, male choice for larger females, or mutual choice, but does not distinguish among them. The negative hind-leg gradients and male response surface additionally suggest testable hypotheses involving movement, leverage, stability while mounted, or attachment endurance.

313 Pair duration and sexual conflict are especially relevant alternatives. In the seed beetle
314 *Callosobruchus maculatus*, spined male genitalia damage the female reproductive tract and
315 female kicking shortens copulation and reduces damage (Crudgington & Siva-Jothy, 2000). I did
316 not quantify attachment duration in *Lyperobates* sp. A or collect systematic observations of
317 kicking, induced detachment, genital morphology, or reproductive-tract scarring. A phenotype
318 associated with prolonged attachment could therefore be overrepresented among observed pairs
319 even without increasing the number of matings or fertilizations. Direct observations of pair
320 formation, detachment, remating, and copulation duration would distinguish mate acquisition
321 from persistence and sexual conflict.

322 The qualitative similarity of the directional gradients between the sexes is a central feature of the
323 results. Both females and males showed positive body-length gradients and negative hind-leg
324 gradients, with similar estimated magnitudes. Although the male analysis supported a
325 correlational gradient and the corresponding female gradient was not supported, the pooled sex
326 comparison provided little evidence that the cross-product association differed between females
327 and males. The apparent male-only result may therefore reflect lower precision in the female
328 analysis rather than a genuinely sex-specific surface. Existing dimorphism could instead reflect
329 past selection, sex-specific viability, development, ecology, genetic architecture, or fitness
330 components not measured here (Blanckenhorn, 2000; Fairbairn, 1997; Fairbairn et al., 2007).

331 Inbreeding is another potential modifier: it can depress male morphology and reproduction
332 (Vasudeva et al., 2025), but relatedness and genetic load were not measured in this population
333 and cannot be inferred from the restricted spatial sample. The greater proportional dimorphism in
334 body length than in hind leg length likewise does not map straightforwardly onto the
335 contemporary gradients, and there was no clear sex difference in the static allometric slope

relating the two traits. This mismatch illustrates why a single fitness component cannot by itself identify the historical cause of SSD.

The absence of body-length assortment indicates that, among the 25 pairs sampled, large males were no more likely to be paired with large females than expected by chance. This result does not contradict selection through pairing status. Morphology may affect the probability of being paired while having little effect on the phenotype of the eventual partner (McDonald & Pizzari, 2016). The modest number of pairs nevertheless limits power to detect weak assortment, and a single nocturnal snapshot cannot reveal whether assortment varies through the season or across sites.

The main inferential limit is the cross-sectional design: 25 pairs were sampled at one site during 11 nights, egg load was measured once, and pairing behaviour was not recorded directly. Pairing status is consequently a measure of observed precopulatory success, not lifetime mating or reproductive success, and the sampled sex ratio may reflect activity or detectability rather than abundance. The study nevertheless provides comparative field estimates for both sexes of a poorly known species. Replication across dates, populations, and related weevils, combined with observations or experiments on mate searching, choice, attachment, and reproductive output, would test the mechanisms suggested by the present patterns.

In conclusion, female and male *Lyperobates* sp. A paired nonrandomly with respect to size: longer-bodied, shorter-legged individuals of both sexes were more likely to be observed paired. Larger females also carried more eggs, and females were larger than males in both traits. The male analysis further identified a high-pairing region combining large bodies with short legs, although the pooled comparison did not establish that the correlational association differed between the sexes. Size therefore matters for both female reproductive investment and pairing in

359 this wild population, while the correspondence between contemporary selection and existing
360 female-biased dimorphism remains incomplete.

361

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460

Table 1: Selection differentials (S, C) and standardized linear (β) and diagonal quadratic (γ) selection gradients through observed pairing status for female and male *Lyperobates* sp. A weevils. Estimates are means $\pm$ SE. Differentials quantify total phenotypic selection, whereas gradients describe associations with each trait after accounting for the other trait. Estimates were returned directly by `gsg::gam.gradients()` on the relative-fitness scale; diagonal quadratic gradients and SEs were not doubled. Standard errors and two-sided *P*-values were obtained from 10,000 parametric-bootstrap replicates.

Trait	Differentials				Gradients			
	$S_i \pm \text{SE}$	<i>P</i>	$C_{ii} \pm \text{SE}$	<i>P</i>	$\beta_i \pm \text{SE}$	<i>P</i>	$\gamma \pm \text{SE}$	<i>P</i>
(a) Females								
body length	-0.040 $\pm$ 0.146	0.788	-0.388 $\pm$ 0.220	0.090	0.930 $\pm$ 0.391	0.032	-0.593 $\pm$ 2.163	0.391
hind leg length	-0.148 $\pm$ 0.153	0.335	-0.246 $\pm$ 0.217	0.283	-1.224 $\pm$ 0.332	<0.001	0.454 $\pm$ 0.406	0.105
body length $\times$ hind leg length							-0.228 $\pm$ 0.483	0.414
(b) Males								
body length	0.210 $\pm$ 0.165	0.205	-0.092 $\pm$ 0.199	0.620	0.968 $\pm$ 0.247	<0.001	0.601 $\pm$ 0.354	<0.001
hind leg length	-0.175 $\pm$ 0.165	0.289	-0.121 $\pm$ 0.213	0.553	-0.954 $\pm$ 0.251	<0.001	0.584 $\pm$ 0.360	<0.001
body length $\times$ hind leg length							-0.593 $\pm$ 0.335	<0.001

Figure 1: Mating pair of *Lyperobates* sp. A weevils. The male is mounted dorsally on the female in a typical copulatory position. (a) Lateral (side) view showing the male positioned atop the female's abdomen, with legs grasping the female's body. (b) Frontal view illustrating the alignment of the pair and relative positioning of the male on the female's dorsum.

461

Figure 2: Sexual dimorphism in body and hind leg length. Boxplots show total body length and total hind leg length (mm) for females (light grey) and males (dark grey), with points representing individuals. Horizontal lines are medians, boxes are interquartile ranges, and whiskers extend to 1.5× the interquartile range. Females were significantly larger than males for both traits, as assessed by the linear models reported in the Results.

462

Figure 3: Relationship between female body length and egg load. Points represent individual females, and egg load is the number of eggs present in the reproductive tract at dissection. Body length is plotted on a log scale. The line shows the fitted mean from the negative-binomial generalized linear model, and the shaded region is the 95% confidence interval. Egg load increased with body length; this cross-sectional measure is not equivalent to lifetime realized fecundity.

463

Figure 4: Associations between morphology and observed pairing status in males (a–b) and females (c–d). Panels show the predicted probability of being observed paired from binomial generalized additive models as a function of body length (a, c) and hind leg length (b, d), averaged over the observed distribution of the alternate trait. Solid lines are model predictions and dashed lines are 95% intervals estimated by parametric bootstrapping. In both sexes, body length had a positive directional gradient and hind leg length had a negative directional

gradient. Pairing status is a cross-sectional proxy and may reflect encounter, mounting, or pair duration rather than lifetime reproductive success.

464

Figure 5: Predicted probability that a male was observed paired as a function of body length and leg length. Colours and labelled contours represent fitted probabilities from the generalized additive model. White circles show unpaired males and yellow triangles show paired males; predictions are restricted to the convex hull of the observed combinations of body and leg length. Higher pairing probabilities were associated with greater body length relative to leg length.

465

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2

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Figure 2

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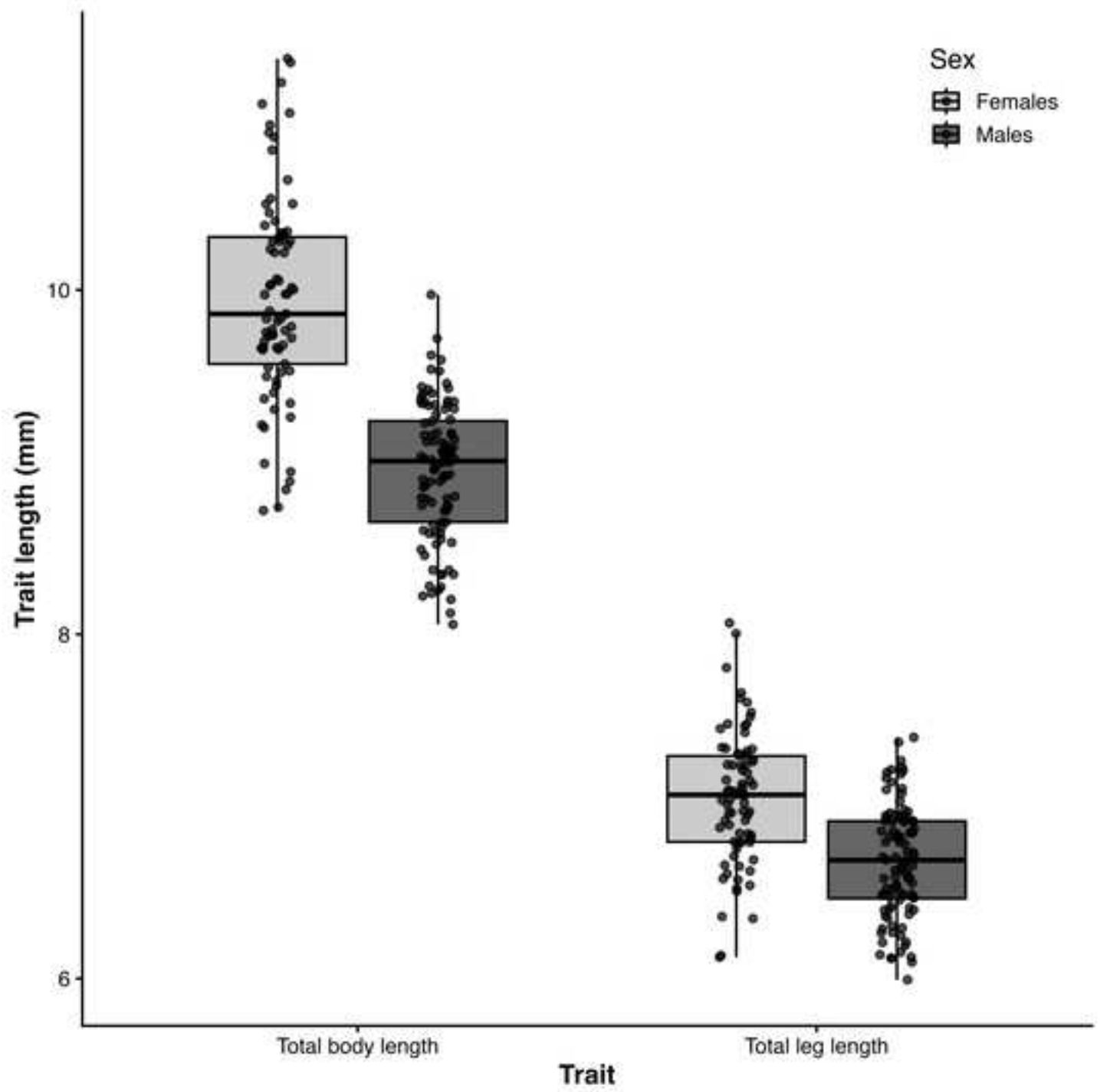
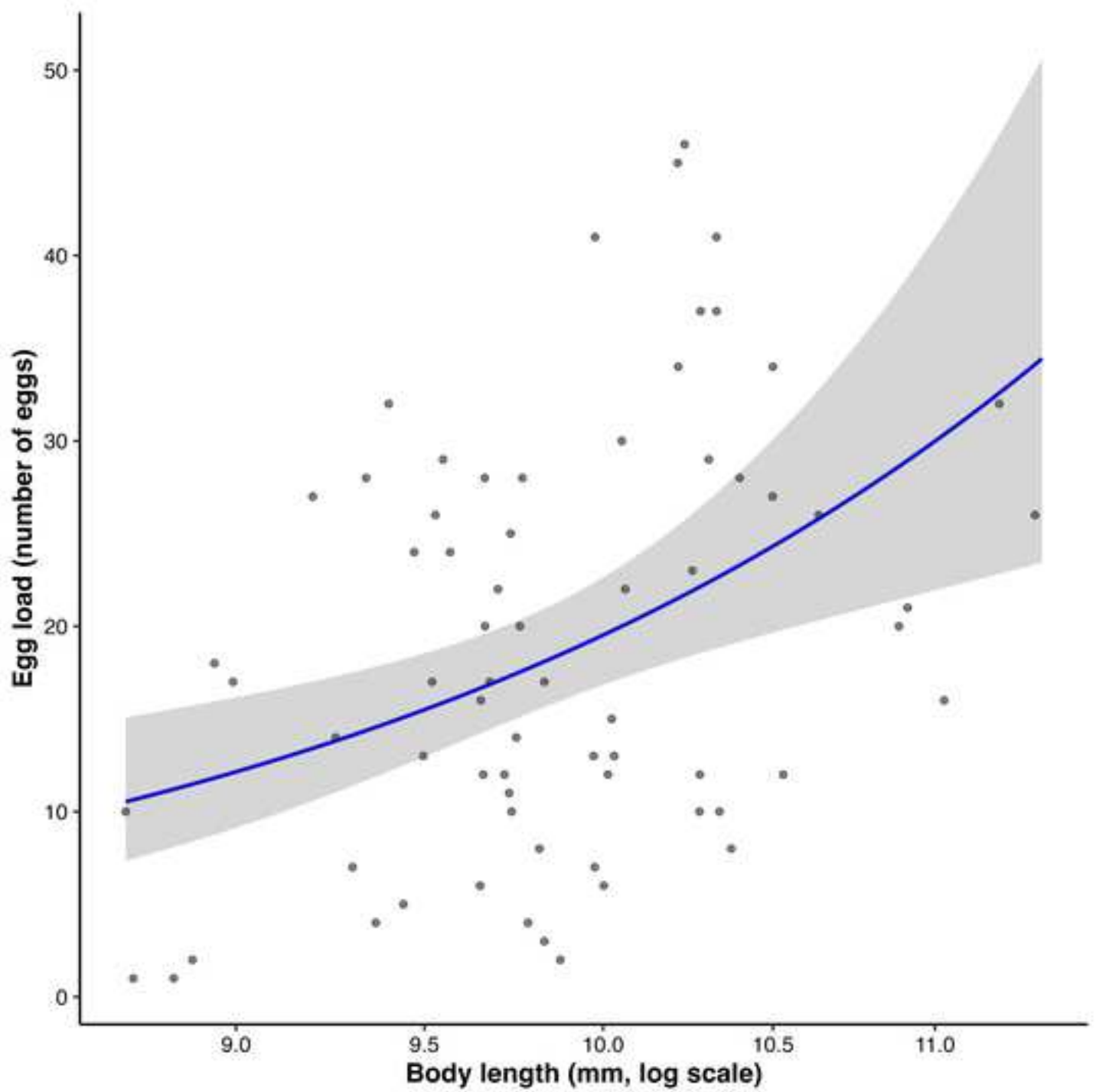


Figure 3

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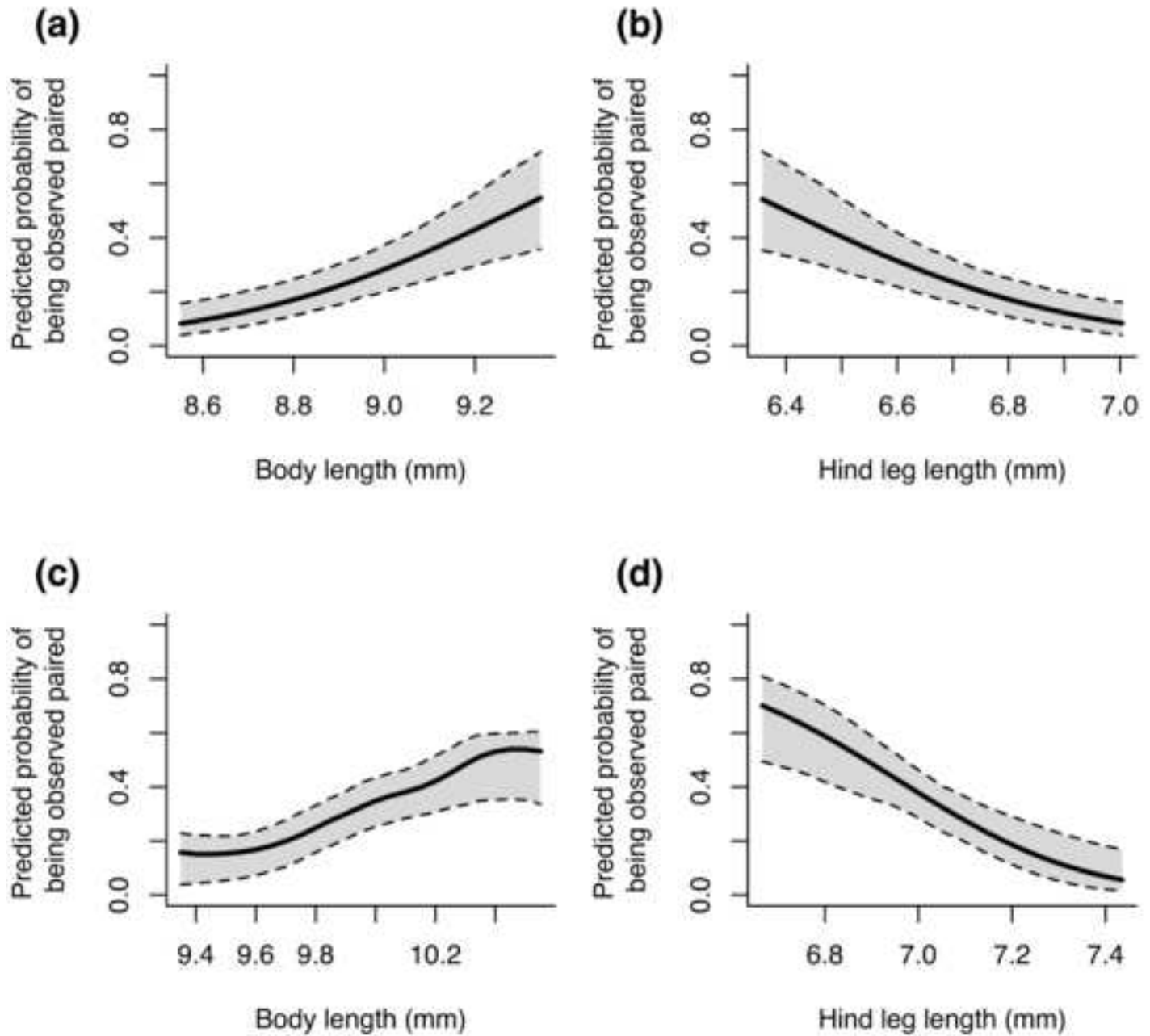


Figure 5

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